

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIC504 - CONCRETE TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following physical properties of aggregates most influences the bond between aggregate and cement paste?
 - a) Water absorption
 - b) Shape and surface texture
 - c) Aggregate impact value
 - d) Bulk density
- 2 Which of the following statements about C₃A (Tricalcium Aluminate) is correct?
 - a) It contributes significantly to long-term strength
 - b) It hydrates slowly and reduces heat of hydration
 - c) It reacts rapidly and is controlled by adding gypsum
 - d) It is responsible for high soundness of cement
- 3 Which transportation method is most likely to cause loss of workability due to continuous agitation and evaporation?
 - a) Pumping
 - b) Belt conveyors
 - c) Truck mixers over long distances
 - d) Wheelbarrow transport at site
- 4 Bleeding in fresh concrete is most likely when:
 - a) Cement content is very high
 - b) Fine aggregate content is low and W/C ratio is high
 - c) Concrete is heavily vibrated
 - d) Curing starts immediately after finishing

- 5 Air-entraining agents are added to concrete mainly to:
 - a) Increase density
 - b) Improve freeze–thaw resistance
 - c) Reduce setting time
 - d) Increase compressive strength

- 6 Accelerators in concrete are useful when:
 - a) Concrete is placed in hot climates
 - b) Faster formwork removal or early strength is needed
 - c) Concrete must remain plastic for longer time
 - d) Concrete with high slump is required

- 7 Characteristic strength of concrete (f_{ck}) is defined as:
 - a) Mean strength of concrete at 28 days
 - b) Strength exceeded by 95% of test results
 - c) Strength below which 5% of test results are expected to fall
 - d) Maximum compressive strength obtained in the batch

- 8 Chloride-induced corrosion of reinforcement occurs mainly because chlorides:
 - a) Reduce cement fineness
 - b) Break the passive oxide film on steel surface
 - c) Increase setting time of concrete
 - d) Reduce elastic modulus of concrete

- 9 Flexural strength of concrete (modulus of rupture) is generally:
 - a) 0.1 to 0.2 times its compressive strength
 - b) 0.7 to 0.8 times its compressive strength
 - c) 3 to 4 times its compressive strength
 - d) Equal to its tensile strength

- 10 In concrete mix design, the water–cement ratio is primarily selected based on:
 - a) Desired slump
 - b) Nominal mix proportions
 - c) Required strength and durability
 - d) Specific gravity of aggregates

- 11 In mix proportioning, the aggregate ratio and grading affect concrete mainly by controlling:
 - a) Heat of hydration
 - b) Workability and void content
 - c) Setting time
 - d) Carbonation depth

- 12 A “trial mix” is prepared to:
 - a) Verify only the slump
 - b) Adjust material proportions to meet strength and workability requirements
 - c) Replace cube testing
 - d) Check curing performance

Q2. Solve the following.

- A) What is hydration of cement? Explain the hydration mechanism and heat of hydration.

6

- B) What are the important physical properties of aggregates required for good concrete? 6
- Q3. Solve the following.
- A) Define workability. Explain the factors affecting workability of concrete. 6
- B) Describe different methods of curing concrete with neat sketches. 6
- Q4. Solve Any Two of the following.
- A) Define admixtures and explain their role in modifying the properties of concrete. 6
- B) Explain the effect of plasticizers on: a) Workability b) Setting time 6
- C) Describe the environmental and economic advantages of green concrete. 6
- Q5. Solve Any Two of the following.
- A) Describe the factors affecting the compressive strength of concrete. 6
- B) Explain creep and shrinkage with its significant in accordance with durability of concrete. 6
- C) What is alkali aggregate reaction? State the factors affecting on alkali aggregate reaction. 6
- Q6. Solve Any Two of the following.
- A) What do you mean by concrete mix design? What are the objectives in mix design? 6
- B) Define nominal mix concrete. Give examples of commonly used nominal mixes as per IS 456. 6
- C) Define non-destructive testing of concrete and state its importance. Explain in short Ultrasonic pulse velocity (UPV) test. 6

***** End *****